COMPUTER VISION PROJECT REPORT

Efficient Image Clustering Using SIFT + CNN Ensemble Features

Caltech-101 Subset · Bag of Visual Words · PCA · MobileNetV2 · Feature Fusion

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A comparative study of handcrafted local descriptors, pretrained deep visual representations, and hybrid feature-fusion strategies.

Abstract

This project investigates unsupervised image clustering by comparing handcrafted visual features, deep convolutional features, and combinations of the two. A 604-image subset of the Caltech-101 dataset, covering ten object categories, was represented using (i) Scale-Invariant Feature Transform (SIFT) descriptors aggregated into a Bag of Visual Words (BoVW) histogram and reduced with Principal Component Analysis (PCA), and (ii) 1,280-dimensional features extracted from an ImageNet-pretrained MobileNetV2 network. The feature representations were aligned to a common 50-dimensional space and combined using averaging, weighted, power/gamma, logarithmic, exponential, and geometric-mean fusion. K-Means with ten clusters was used, and clustering was evaluated using Silhouette, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI). The supplied presentation reports that deep features substantially outperform SIFT alone and that selected fusion methods improve the reported semantic agreement with the dataset labels. The report also identifies a labelling inconsistency between the presentation and notebook outputs that should be resolved before formal submission.

Executive summary

The main finding is that the pretrained CNN representation carries substantially stronger category information than the SIFT-BoVW representation in this experiment. The presentation reports an ARI of 0.7827 and NMI of 0.8941 for MobileNetV2, compared with 0.0378 and 0.0946 for SIFT. It further reports strong results for averaging and CNN-dominant weighted fusion. However, the notebook output order appears to associate the two principal fusion result rows with the opposite fusion labels. This is a material reporting issue: the numerical results are available, but the labels must be checked against a clean rerun before the report is used as a final assessed submission.

Key reported outcomes

Finding	Reported result
Dataset	604 images across 10 Caltech-101 categories
SIFT baseline	ARI 0.0378; NMI 0.0946; Silhouette –0.0122
MobileNetV2 baseline	ARI 0.7827; NMI 0.8941; Silhouette 0.0756
Best reported ARI/NMI	Weighted fusion row: ARI 0.9100; NMI 0.9513
Highest reported Silhouette among the main methods	Averaging row: 0.1911 (excluding exponential fusion)

Interpretation: Silhouette measures geometric separation without using the true labels; ARI and NMI compare the resulting clusters with the known category labels for evaluation only.

1. Introduction

Image clustering groups visually similar images without requiring class labels during the clustering process. It is useful when large image collections must be organised, explored, or indexed and manual annotation is expensive. The quality of a clustering result depends strongly on how each image is represented: a feature representation should preserve visual similarities that are meaningful for the task.

This project compares two complementary representation families. SIFT describes local keypoint patterns such as corners and texture structures. MobileNetV2 produces higher-level feature vectors learned from a large image-recognition dataset. The project then investigates whether combining local handcrafted information with deep semantic information can improve clustering.

1.1 Problem statement

The project addresses the following question: **Can combining SIFT-based local features with pretrained MobileNetV2 features improve unsupervised image clustering compared with using either representation alone?**

1.2 Aim and objectives

The aim is to evaluate image-clustering quality across handcrafted, deep-learning, and hybrid feature representations. The specific objectives are:

- Construct a ten-category subset of Caltech-101 and prepare a common image collection.
- Extract SIFT descriptors and build a 200-word visual vocabulary using MiniBatch K-Means.
- Extract 1,280-dimensional image embeddings using pretrained MobileNetV2.
- Reduce and align feature representations to 50 dimensions for fusion.
- Compare multiple fusion functions and cluster the resulting representations using K-Means.
- Evaluate clustering with Silhouette, ARI, and NMI, and inspect two-dimensional PCA plots.

1.3 Scope

The experiment focuses on ten selected Caltech-101 categories and a fixed clustering count of ten. The supplied materials do not report a runtime or memory benchmark; therefore, the word “efficient” in the project title is treated as the project framing rather than a measured efficiency claim.

2. Dataset and preprocessing

2.1 Dataset

The source dataset is Caltech-101, which contains 102 categories. The experiment uses the following ten categories: dolphin, minaret, trilobite, lamp, water_lilly, panda, wheelchair, buddha, crocodile, and ceiling_fan. Up to 50 images were selected per category, resulting in 604 images in total. The number is below 500? No: the actual total is 604, reflecting the availability of images in the selected categories.



Figure 1. Sample images from the selected Caltech-101 subset (notebook output).

2.2 Dataset characteristics

The selected categories contain animals, objects, and structures. This provides a varied set of shapes, textures, and global appearances. Images within a category can vary substantially, while different categories may share visual characteristics. These properties make the subset a useful test for comparing local descriptors and semantic feature representations.

2.3 Sampling and reproducibility

The notebook samples ten category folders using Python's random.sample and does not set a random seed for category selection or image selection. The saved notebook output records the ten categories listed above and 604 images. A fresh run may select a different subset. For a reproducible experiment, the selected category list and image filenames should be stored explicitly, or the random seed should be fixed and documented.

3. Methodology

The implementation is organised into feature extraction, dimensionality reduction, feature fusion, clustering, and evaluation. The same selected image collection is intended to be used across all pipelines.

3.1 SIFT feature extraction

SIFT (Scale-Invariant Feature Transform) detects local keypoints and computes a 128-dimensional descriptor for each detected keypoint. The notebook extracts descriptors from grayscale versions of the images. Across the 604 images, the saved run reports 257,847 descriptors, each with 128 values.

3.2 Bag of Visual Words

A visual vocabulary is learned by clustering the local descriptors with MiniBatch K-Means using 200 clusters. Each visual-word cluster acts as a visual token. Each image is represented by a 200-bin histogram that counts the visual words found in that image. The histograms are L1-normalised, producing a feature matrix of shape 604 × 200.

3.3 PCA on SIFT features

PCA reduces the 200-dimensional BoVW histograms to 50 dimensions. The resulting SIFT representation is standardised before clustering and fusion. The notebook records a final SIFT feature matrix of 604 × 50.

3.4 MobileNetV2 feature extraction

MobileNetV2 is loaded with ImageNet pretrained weights, with the classification top removed and global average pooling enabled. Images are resized to 224 × 224 pixels and passed through the model's preprocessing function. Each image produces a 1,280-dimensional feature vector, giving a reported CNN feature matrix of 604 × 1,280. The features are then standardised.

3.5 Dimensionality alignment

To combine the CNN and SIFT representations, PCA is applied to the standardised CNN features to reduce them from 1,280 to 50 dimensions. This makes both feature sets 50-dimensional, allowing element-wise fusion. The notebook uses the 50-dimensional SIFT-PCA representation and the 50-dimensional CNN-PCA representation for the fusion experiments.

3.6 Feature-fusion strategies

The project tests several element-wise fusion operations. Let s denote the 50-dimensional SIFT representation and c denote the 50-dimensional CNN representation after the described scaling and dimensionality alignment.

Method	Definition / setting	Purpose
Averaging	f = 0.5(s + c)	Treats both feature sources equally.
Weighted	f = 0.2s + 0.8c (α = 0.8)	Gives greater influence to CNN features.
Power / Gamma	f = (s + shift)^0.8 ■ (c + shift)^0.2	Applies a nonlinear element-wise combination.
Log	log(1 + s + shift) + log(1 + c + shift)	Compresses feature magnitudes after shifting.
Exponential	exp(2(s + c))	Tests a strongly magnitude-amplifying transform.
Geometric mean	sqrt((s + shift)(c + shift) + ε)	Emphasises dimensions with joint support.

The code applies small positive shifts where required to handle negative values before power, logarithmic, or geometric operations. Exponential and geometric-mean outputs are standardised before clustering in the notebook.

3.7 Clustering algorithm

K-Means is used with k = 10, matching the number of selected categories. K-Means assigns each feature vector to the nearest cluster centroid and iteratively updates centroids to reduce within-cluster squared distances. The

implementation uses random_state=42 and n_init='auto'.

3.8 Evaluation metrics

Silhouette score: Measures how close each sample is to its own cluster compared with the nearest alternative cluster. Values closer to +1 generally indicate better separation; values near zero indicate overlap; negative values suggest possible misassignment.

Adjusted Rand Index (ARI): Measures agreement between predicted clusters and known class labels, adjusted for chance. A value near 1 indicates strong agreement; a value near 0 is close to chance-level agreement.

Normalized Mutual Information (NMI): Measures the shared information between cluster assignments and the true labels, normalised to a comparable scale. Higher values indicate stronger correspondence.

The true labels are used only to calculate ARI and NMI after clustering; they are not used to train the feature extractors or K-Means clustering.

4. Experimental results

The following tables reproduce the metric values shown in the supplied project presentation. They should be interpreted alongside the notebook-label verification note in Section 6.

Representation / fusion	Silhouette	ARI	NMI
SIFT (BoVW + PCA)	–0.0122	0.0378	0.0946
MobileNetV2 CNN	0.0756	0.7827	0.8941
Averaging (presentation label)	0.1911	0.8801	0.9401
Weighted, $\alpha = 0.8$ (presentation label)	0.1702	0.9100	0.9513
Power / Gamma, $\gamma = 0.8$	0.1205	0.8640	0.9261
Log fusion	0.1552	0.7712	0.8880
Exponential, $\beta = 2.0$	0.8377	0.0012	0.0307
Geometric mean	0.0246	0.2957	0.5746

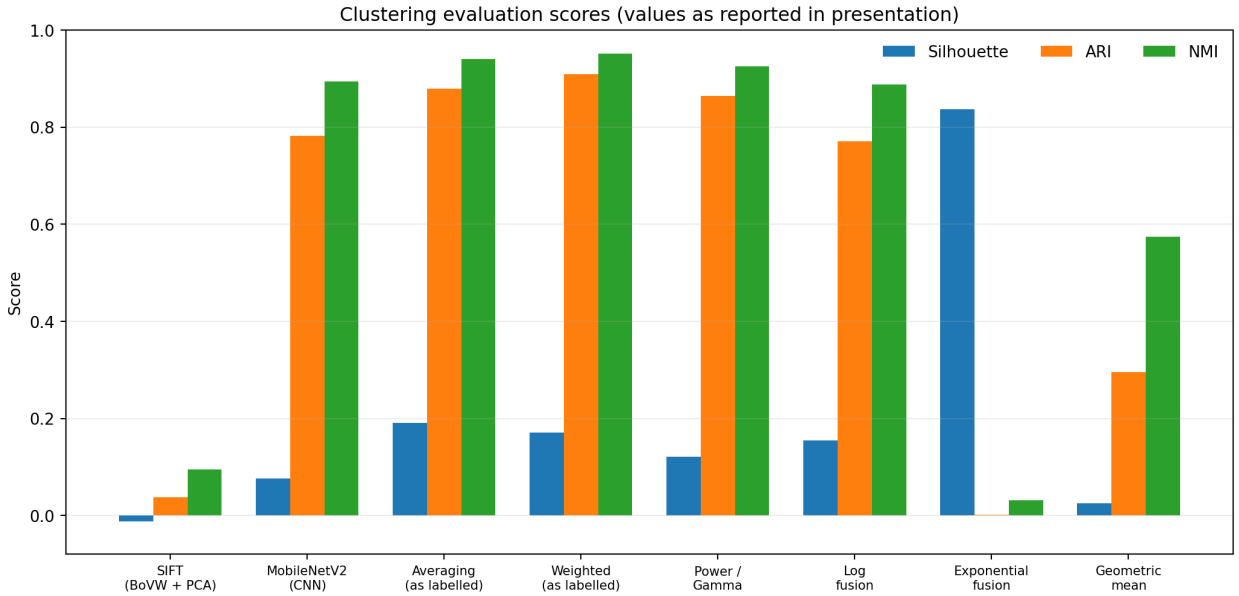


Figure 2. Comparison of the reported Silhouette, ARI and NMI scores. The exponential method's high Silhouette score contrasts sharply with its very low ARI and NMI.

4.1 Baseline comparison

SIFT alone performs poorly against the known categories: its Silhouette score is slightly negative, and both ARI and NMI are close to zero. This suggests that the local visual-word histograms, in this configuration, do not provide a sufficiently useful representation for separating the ten semantic categories.

MobileNetV2 performs much better on ARI and NMI. The result is consistent with the idea that pretrained deep features capture higher-level object structure and category-relevant patterns that are not represented as effectively by the handcrafted SIFT-BoVW histogram.

4.2 Fusion comparison

The presentation reports that both averaging and CNN-dominant weighted fusion outperform the CNN baseline on ARI and NMI. The weighted row has the highest reported ARI (0.9100) and NMI (0.9513), while the averaging row has the highest Silhouette score among the principal fusion methods (0.1911). This suggests a trade-off between geometric cluster separation and agreement with the known categories.

4.3 Nonlinear fusion observations

Power/gamma fusion and log fusion retain relatively strong ARI and NMI in the reported results, although neither exceeds the best main fusion row. The exponential method produces a very high Silhouette score (0.8377) but almost no agreement with the true categories (ARI 0.0012; NMI 0.0307). This is an important example of why an unsupervised geometric metric should not be interpreted alone: a representation can form compact, separated clusters that do not correspond to the semantic classes of interest.

The geometric mean produces modest agreement (ARI 0.2957; NMI 0.5746), lower than the CNN baseline and the main fusion methods. In this experiment, its element-wise agreement constraint appears less effective than averaging or weighted combination.

5. Visual inspection

The notebook generates two-dimensional PCA projections for the SIFT, CNN, and fused feature spaces. These plots are qualitative diagnostics: PCA preserves directions of greatest variance, not necessarily the dimensions that best separate semantic categories. The visualisations should therefore support, rather than replace, the numerical metrics.

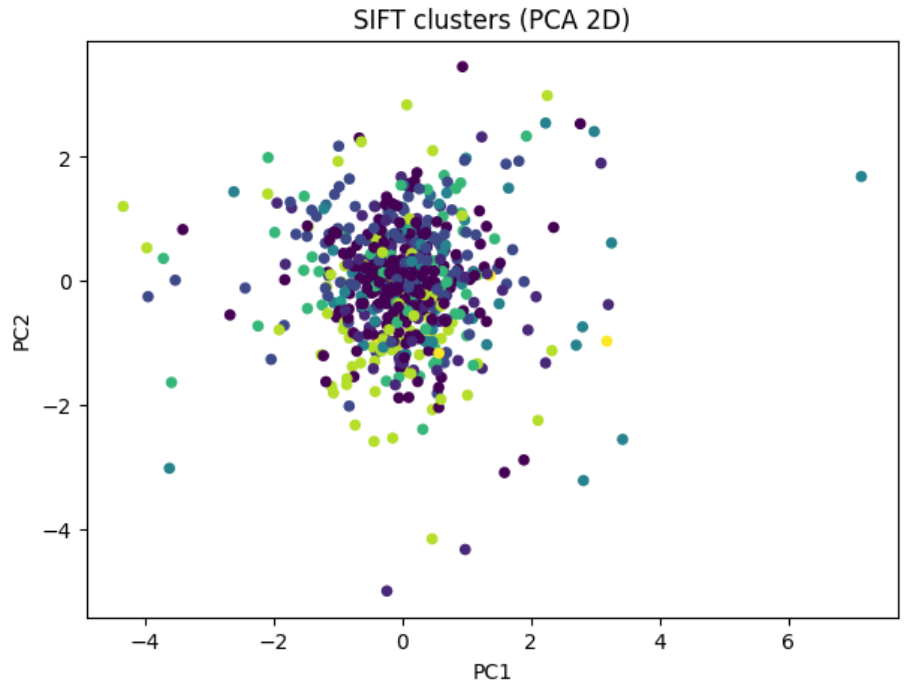


Figure 3. SIFT cluster assignments projected into two PCA dimensions.

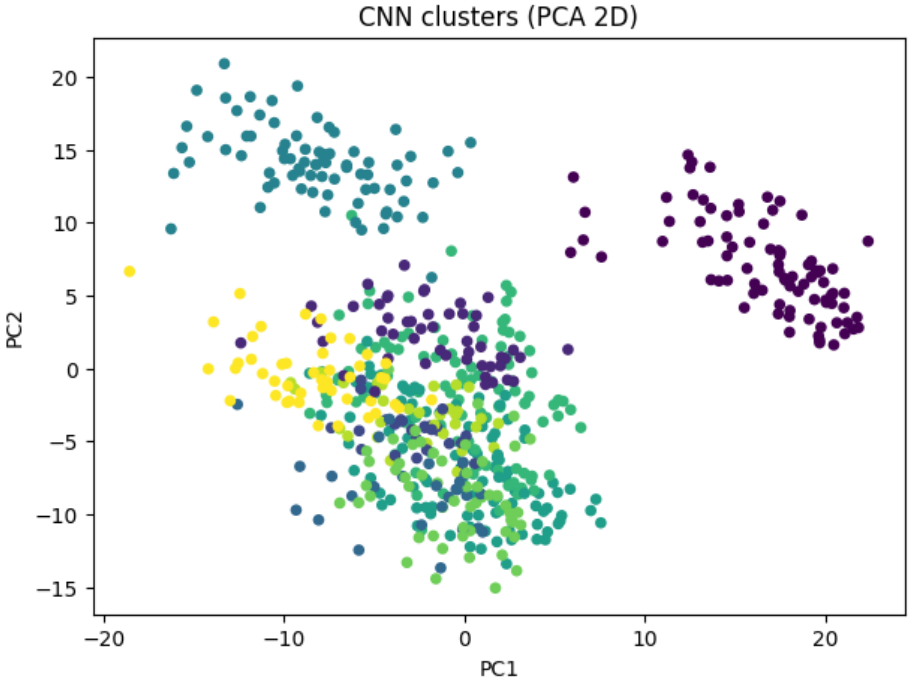


Figure 4. MobileNetV2 cluster assignments projected into two PCA dimensions.

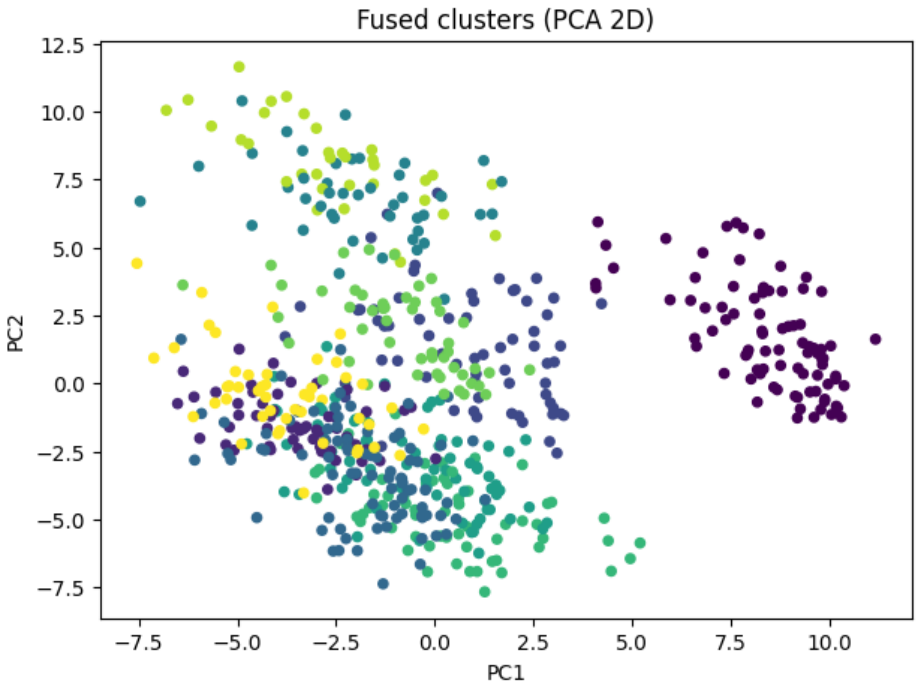


Figure 5. Fused-feature cluster assignments projected into two PCA dimensions.

The supplied presentation describes the SIFT projection as showing substantial overlap, the CNN projection as showing clearer semantic structure, and the fusion projection as showing the strongest grouping. The notebook also displays sample images from selected fused clusters and cluster-size distributions. Such inspection can reveal whether clusters are dominated by coherent categories or contain mixtures.

6. Discussion, validation and limitations

6.1 Main interpretation

The experiment supports three practical conclusions. First, feature representation is central to clustering quality: the pretrained MobileNetV2 representation is much more informative than SIFT-BoVW for this subset. Second, combining complementary representations can improve agreement with category labels. Third, fusion operations must be selected and evaluated carefully; aggressive transformations may create geometrically separated clusters without preserving semantic meaning.

6.2 Important result-label inconsistency to resolve

The presentation labels the 0.1911 / 0.8801 / 0.9401 row as Averaging and the 0.1702 / 0.9100 / 0.9513 row as Weighted fusion. However, in the supplied notebook, the first clustering cell after both fusion matrices are created calls cluster_and_report on X_fused_weighted, and its saved output is 0.1911 / 0.8801 / 0.9401. The following cell calls cluster_and_report on X_fused_avg, and its saved output is 0.1702 / 0.9100 / 0.9513. Therefore, the notebook's execution order appears to associate those two result sets with the opposite fusion labels from the presentation. This report preserves the presentation's labels in the main results table but flags the discrepancy rather than silently choosing one.

Recommended correction: rerun the notebook from a clean runtime, print each fusion method's name immediately beside its metrics, and update both the presentation and report from that single verified run. Ensure the plotted fused cluster assignments also correspond to the method being described.

6.3 Reproducibility and implementation checks

• **Random subset selection:** The notebook does not set a seed before selecting category folders and image files. The saved run has 604 images and the ten categories stated in this report, but another run may differ.
• **Feature/label alignment:** The notebook checks that CNN paths match CNN features, but creates the label vector

from all_labels. It should build the CNN evaluation labels from cnn_labels and verify that the image paths, feature rows, and labels have identical ordering. They match in the saved run (604 items), but this should be asserted explicitly.

• **Cluster initialisation:** K-Means uses a fixed random_state, but the feature subset and upstream MiniBatchKMeans vocabulary can still vary because sampling and vocabulary fitting are not fully controlled by a single reproducibility setup.

• **Metric interpretation:** ARI and NMI are valid for post-clustering evaluation, but should not be used to tune the unsupervised pipeline on a held-out test set unless a separate evaluation protocol is established.

• **Efficiency claim:** The materials do not provide extraction time, clustering time, peak memory, or a comparison of computational cost. Efficiency has not therefore been quantitatively demonstrated.

6.4 Limitations

The experiment uses one subset of ten classes and one principal clustering algorithm, K-Means. The number of clusters is fixed to the number of known categories. The study does not report repeated runs, confidence intervals, sensitivity analysis for vocabulary size or PCA dimensions, or tests on a second dataset. MobileNetV2 is used as a fixed pretrained feature extractor rather than fine-tuned for Caltech-101. These choices are appropriate for a comparative project prototype, but limit the extent to which the findings can be generalised.

7. Future work

The supplied presentation proposes several useful extensions. These can be organised into methodological, representation, and evaluation improvements:

1. **Compare clustering algorithms:** Evaluate Spectral Clustering, Agglomerative Clustering, and DBSCAN alongside K-Means. This will show whether the results depend on the spherical-cluster assumptions of K-Means.
2. **Improve visualisation:** Use UMAP or t-SNE as complementary nonlinear visualisation tools, while retaining numerical metrics for evaluation.
3. **Learn fusion weights:** Replace manually selected weights with trainable fusion or metric-learning approaches, using a clearly separated validation protocol.
4. **Run repeated experiments:** Fix and record seeds, repeat clustering across multiple initialisations, and report mean and standard deviation.
5. **Expand evaluation:** Test more categories and another image dataset; inspect per-class cluster purity and confusion patterns.
6. **Measure efficiency:** Record feature-extraction time, clustering time, memory consumption, and the effect of image count and feature dimension.

8. Conclusion

This project compared handcrafted SIFT-BoVW features, pretrained MobileNetV2 features, and several feature-fusion strategies for unsupervised clustering of a 604-image, ten-category Caltech-101 subset. The supplied results show that SIFT alone provides weak semantic grouping, whereas MobileNetV2 features produce much stronger agreement with the known categories. The presentation further reports that averaging and CNN-dominant weighted fusion improve the ARI and NMI scores, with weighted fusion attaining the highest reported semantic agreement and averaging attaining the highest Silhouette score among the principal methods.

The experiment also demonstrates that a high Silhouette score does not necessarily mean that clusters reflect the desired semantic categories: exponential fusion has the highest reported Silhouette score but near-zero ARI and NMI. The overall conclusion is that deep features provide a strong foundation for this task and that carefully designed fusion can add value. Before final academic submission, the fusion-label inconsistency between the presentation and notebook should be resolved through a clean, reproducible rerun.

References

1. Lowe, D. G. (2004). Distinctive Image Features from Scale-Invariant Keypoints. *International Journal of Computer Vision*.
2. Xie, J., Girshick, R., & Farhadi, A. (2016). Unsupervised Deep Feature Learning for Image Clustering. *Proceedings of the International Conference on Machine Learning (ICML)*.
3. Razavian, A. S., Azizpour, H., Sullivan, J., & Carlsson, S. (2014). CNN Features off-the-shelf: an Astounding Baseline for Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*.
4. Caltech-101 dataset, as used in the supplied project notebook and presentation.
5. TensorFlow/Keras MobileNetV2 application implementation, using ImageNet pretrained weights.

Appendix A. Implementation stack

Python; OpenCV SIFT; scikit-learn (MiniBatchKMeans, PCA, StandardScaler, KMeans and clustering metrics); TensorFlow/Keras MobileNetV2; NumPy; Matplotlib; PIL; Google Colab and Google Drive for dataset access and notebook execution.

Appendix B. Reproducibility checklist

- Freeze the ten category names and image filenames.
- Set and record random seeds for Python, NumPy, and scikit-learn where applicable.
- Re-run all notebook cells from a fresh runtime.
- Verify that each feature matrix and its label/path arrays have the same row count and order.
- Reprint metrics with unambiguous fusion method names.
- Update presentation figures and tables from the verified run.
- Record hardware, software versions, runtime, and memory use.